

Commerce XI Maths App

Let's Remember

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 3 - Skewness | Total questions: 14

Question 1 - MCQ · PG-43 · LR-1

Karl Pearson's coefficient is denoted by

Options

- A. Sk_p
- B. Sk_β
- C. S.D.
- D. Q.D.

Correct answer: Sk_p

Explanation and steps

Sk_p denotes Pearsonian skewness.

Question 2 - MCQ · PG-43 · LR-2

Karl Pearson's coefficient equals

Options

- A. $\frac{\text{Mean} - \text{Mode}}{\text{S.D.}}$
- B. $\frac{\text{Mode} - \text{Mean}}{\text{S.D.}}$
- C. $\frac{Q_3 - Q_1}{2}$
- D. $\frac{\text{Mean} - \text{Median}}{\text{Variance}}$

Correct answer: $\frac{\text{Mean} - \text{Mode}}{\text{S.D.}}$

Explanation and steps

This is the primary Pearsonian formula.

Question 3 - MCQ · PG-43 · LR-3

If mode is indeterminate, Pearson's coefficient equals

Options

- A. $3 \left(\frac{\text{Mean} - \text{Median}}{\text{S.D.}} \right)$
- B. $\frac{\text{Mean} - \text{Median}}{\text{S.D.}}$
- C. $3 \left(\frac{\text{Mode} - \text{Median}}{\text{S.D.}} \right)$
- D. $\frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$

Correct answer: $3 \left(\frac{\text{Mean} - \text{Median}}{\text{S.D.}} \right)$

Explanation and steps

Use the empirical mean–median–mode relation.

Question 4 - MCQ · PG-44 · LR-4

For a symmetric distribution, Sk_p is

Options

- A. 0
- B. 1
- C. -1
- D. 3

Correct answer: 0

Explanation and steps

No skewness means coefficient zero.

Question 5 - MCQ · PG-44 · LR-5

For a positively skewed distribution, Sk_p is

Options

- A. greater than 0
- B. less than 0
- C. equal to 0
- D. equal to -1

Correct answer: greater than 0

Explanation and steps

Positive skew gives a positive coefficient.

Question 6 - MCQ · PG-44 · LR-6

For a negatively skewed distribution, Sk_p is

Options

- A. less than 0
- B. greater than 0
- C. equal to 0
- D. equal to 1

Correct answer: less than 0

Explanation and steps

Negative skew gives a negative coefficient.

Question 7 - MCQ · PG-44 · LR-7

Most Sk_p values lie between

Options

- A. -1 and 1
- B. 1 and 3
- C. 0 and 1
- D. -10 and 10

Correct answer: -1 and 1

Explanation and steps

This is the usual range stated in the textbook.

Question 8 - MCQ · PG-44 · LR-8

All Sk_p values lie between

Options

- A. -3 and 3
- B. -1 and 1
- C. 0 and 3
- D. -100 and 100

Correct answer: -3 and 3

Explanation and steps

This is the full textbook range.

Question 9 - MCQ · PG-44 · LR-9

Bowley's coefficient is denoted by

Options

- A. Sk_β
- B. Sk_p
- C. S.D.
- D. C.V.

Correct answer: Sk_β

Explanation and steps

Sk_β denotes Bowley skewness.

Question 10 - MCQ · PG-44 · LR-10

Bowley's coefficient equals

Options

- A. $\frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$
- B. $\frac{Q_3 - Q_1}{2}$
- C. $\frac{\text{Mean} - \text{Mode}}{\text{S.D.}}$
- D. $\frac{Q_3 + Q_1}{2Q_2}$

Correct answer: $\frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$

Explanation and steps

This is Bowley's quartile formula.

Question 11 - MCQ · PG-44 · LR-11

If $Sk_\beta > 0$, the distribution is

Options

- A. positively skewed
- B. negatively skewed
- C. symmetric
- D. without a median

Correct answer: positively skewed

Explanation and steps

The positive sign indicates positive skewness.

Question 12 - MCQ · PG-44 · LR-12

If $Sk_b < 0$, the distribution is

Options

- A. negatively skewed
- B. positively skewed
- C. symmetric
- D. uniform

Correct answer: negatively skewed

Explanation and steps

The negative sign indicates negative skewness.

Question 13 - MCQ · PG-44 · LR-13

Bowley's coefficient lies between

Options

- A. -1 and 1
- B. -3 and 3
- C. 0 and 1 only
- D. -10 and 10

Correct answer: -1 and 1

Explanation and steps

Bowley's coefficient is bounded by -1 and 1 .

Question 14 - MCQ · PG-44 · LR-14

If $Sk_b = 0$, the distribution is

Options

- A. symmetric
- B. positively skewed
- C. negatively skewed
- D. without a median

Correct answer: symmetric

Explanation and steps

A zero Bowley coefficient indicates a symmetric distribution.